

Twitter COVID-19 Network Analysis

Hashtag Co-occurrence, Temporal Decomposition
and Negative-Sentiment Propagation

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Social Network Analysis – Project 13

`github.com/MatusMandzak/sna-project`

Project 13

Build a hashtag-co-occurrence graph from **TweetsCOVID19** (Oct–Nov 2019), characterise it statically and over time, and test whether negative sentiment propagates like an SIS contagion.

Static

- $|V|, |E|$, components, degree centrality
- top-3 connected components
- degree-centrality CDF
- power-law fit + R^2

Temporal

- 10 equal time increments
- per-increment subgraph metrics
- triangle evolution
- sentiment evolution
- SIS contagion calibration

- **TweetsCOV19** part 1
(Oct 2019 – Apr 2020)
- gzip TSV, 820 MB compressed, ~8.15 M lines
- streaming filter on the timestamp column
- kept **287 990** tweets in Oct–Nov 2019 from **188 649** users
- **764 341** hashtag instances over **209 330** unique tags

Hashtag	Tweets
#china	7 134
#healthcare	4 835
#hongkong	4 050
#hongkongprotests	2 785
#strongertogether	2 623
#standwithhongkong	2 563
#epsteincoverup	2 079
#chinazi	1 879
#mamavote	1 833

Pre-pandemic window – #covid19 not yet in the popular tail.

Stages (one Python module each)

- 1 preprocess.py stream-decompress the 820 MB TSV, filter timestamp → tweets_oct_nov_2019.parquet
- 2 build_graph.py invert to *hashtag*→*users*, generate edges → graph.gpickle + adjacency.sqlite
- 3 analysis.py Tasks 2–9 (metrics, components, centrality, power-law, time decomposition, triangles, sentiment, SIS)
- 4 extra_tables.py top-hashtag table + daily volume figure
- 5 polish_tables.py reformat all CSVs into report-ready .tex tables

- End-to-end ~5 min on a consumer laptop (uv run python <stage>).
- Each stage is idempotent and writes only to output/; the report \input-s the polished tables and the figures/*.pdf directly.

Graph construction

- Nodes: anonymised user IDs.
- Edge (u, v) : u and v share at least one hashtag.
- Edge weight w_{uv} = number of shared hashtags.

Tractability cap (deviation 1)

Drop hashtags with > 150 distinct users.

- suppresses star-cliques from #china (3 726 users), #healthcare (2 932), #hongkong (1 769)
- shrinks ~ 39 M raw edges $\rightarrow$ **4.85 M**
- one-line override: `HASHTAG_USER_CAP = float("inf")` restores literal-spec compliance

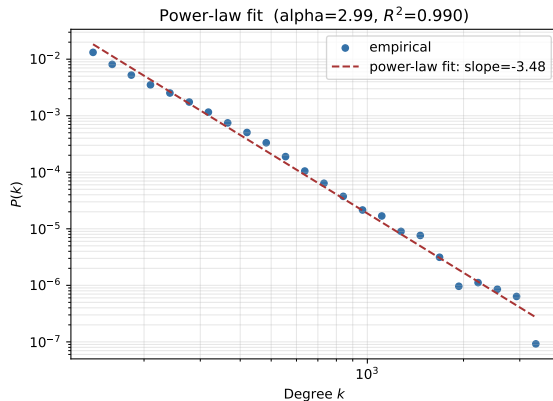
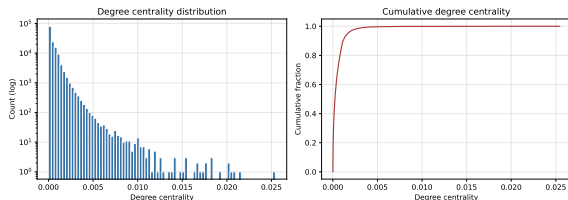
Adjacency A persisted as SQLite (nodes, adjacency).

Global statistics of G

Metric	Value
Nodes	141 562
Edges	4 853 728
Components	2 906
Largest CC size	133 689 (94%)
Avg degree	68.57
Min deg. centrality	0.00001
Max deg. centrality	0.02544
Mean deg. centrality	0.00048
Avg shortest path (sampled)	3.80

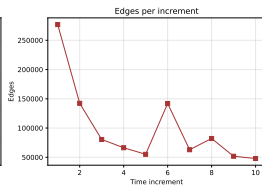
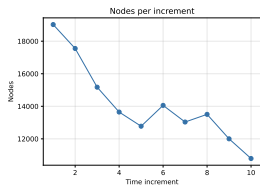
- LCC **absorbs 94 %** of nodes
- small-world: $L \approx 3.8$ hops
- heavy tail: max centrality is $\sim 50\times$ the mean
- 2nd & 3rd components are *exact 50- / 45-cliques* (density 1) — niche hashtag coalitions

Degree centrality & power-law fit



- `powerlaw.Fit` (Clauset–Shalizi–Newman MLE): $\alpha = 2.99$, $x_{\min} = 116$
- log-binned tail regression: slope = -3.48 , $R^2 = 0.99$
- **but** `distribution_compare` prefers *lognormal* over power-law ($R = -10.6$, $p < 10^{-25}$) – consistent with Broido & Clauset 2019.

Temporal decomposition (10 increments)

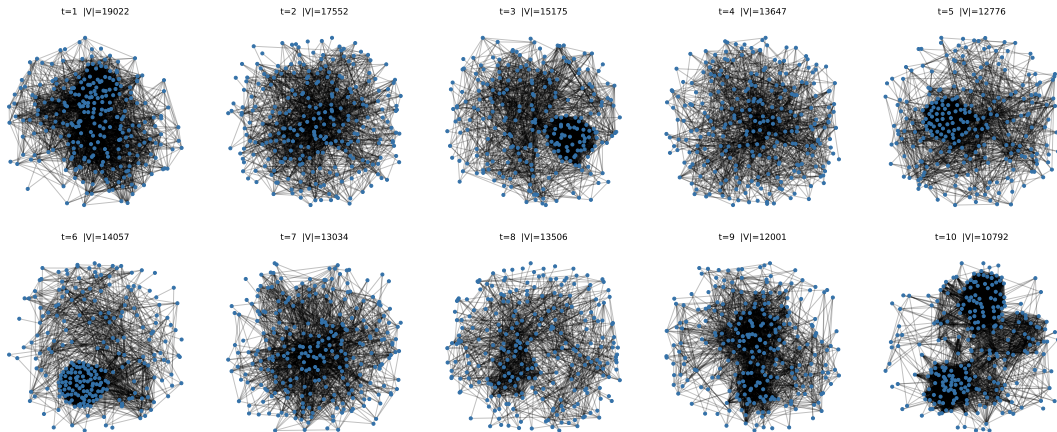


t	$ V $	$ E $	avg path	diam
1	19 022	276 860	3.71	9
2	17 552	142 257	4.56	12
3	15 175	80 482	4.92	11
4	13 647	66 330	5.08	14
5	12 776	55 102	5.38	13
6	14 057	141 915	5.16	13
7	13 034	62 926	5.82	12
8	13 506	82 158	5.88	14
9	12 001	51 806	5.99	14
10	10 792	47 913	6.00	16

Network *shrinks and elongates* over the period; **rebound at $t=6$** aligns with the U.S. impeachment public hearings (mid-November).

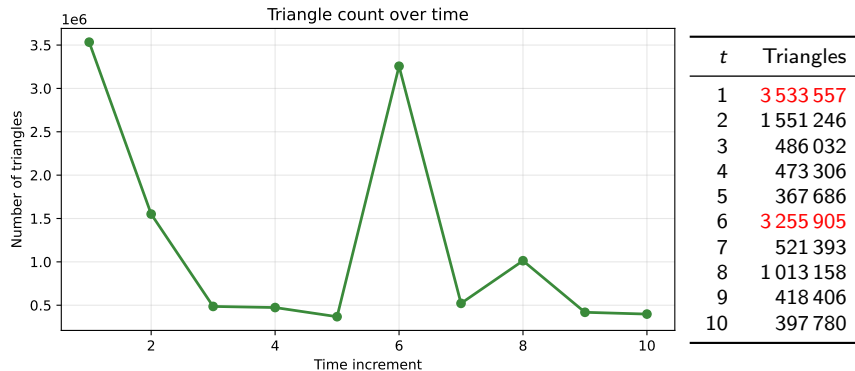
Subgraph LCC drawings

LCC sample of each time increment



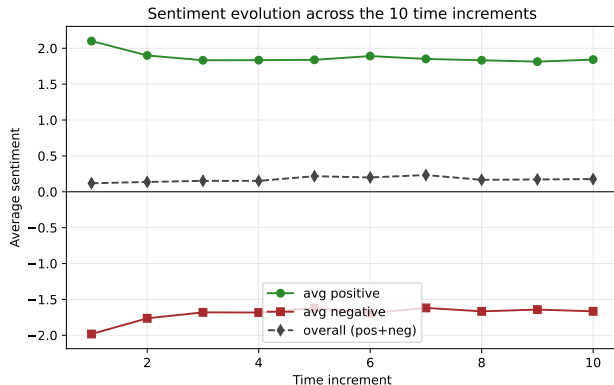
Each panel: largest CC of one increment, BFS-down-sampled to 250 nodes for legibility. Topology drifts from hub-and-spoke towards chain-like as cohorts shrink.

Triangle evolution



October cohorts are densely tri-connected; the $t=6$ rebound mirrors the size rebound and reflects the impeachment-news clustering.

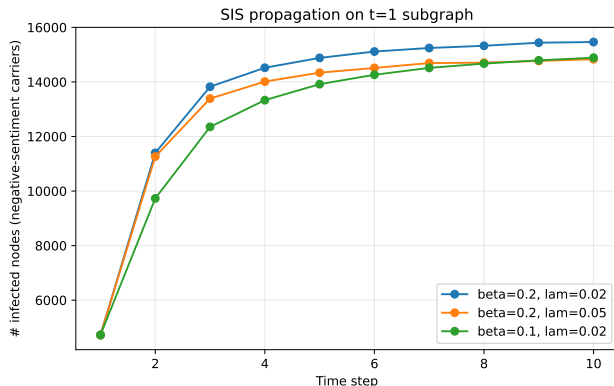
Sentiment evolution



Per-increment aggregation:

$$s = s^+ + s^-, \bar{s}_i = \frac{1}{|V_i|} \sum_{u \in V_i} \sum_{\tau \in T_i(u)} s(\tau)$$

- per-node $\bar{s}^+ \approx +1.9$, $\bar{s}^- \approx -1.7$
- overall sentiment mildly positive ($\sim +0.12$ to $+0.23$)
- $\sum |s_t^-|$ volume drops from $\sim 104k$ ($t=1$) to $\sim 18k$ ($t=10$) – driven by the shrinking active user base.



β	λ	Final	Cumul.
0.20	0.02	15 443	135 754
0.20	0.05	14 852	131 537
0.10	0.02	14 821	126 727
0.20	0.10	13 915	124 692
0.10	0.05	14 146	122 560

- Host: LCC of $t=1$ subgraph (16 036 nodes, 274 k edges)
- target $\sum |s_t^-| = 242\,901$
- best fit reaches $\sim 56\%$
- structural ceiling: cum. infections $\leq 10|V| \approx 1.6 \times 10^5$

Three deliberate departures from the literal spec

1. **150-fanout cap** – otherwise the BFS-bound metrics take hours, not minutes
2. **Sampled** avg path / diameter ($|S| = 200/60/20$) instead of exact NetworkX call
3. **matplotlib drawings** replace “Delphi visualisation” (likely typo for Gephi)

- **Hashtag-projection bias:** co-occurrence $\neq$ information transfer.
- **Power-law caveat:** lognormal favoured formally; we report both honestly.
- **Sentiment $\neq$ pathogen:** news shocks are exogenous, SIS only matches volume, not cause.
- **Pre-pandemic window:** Oct–Nov 2019 is a baseline, not a pandemic snapshot.

What we learned

- G is heavy-tailed, small-world, dominated by a $\sim 94\%$ giant component.
- Power-law $\alpha \approx 3$, $R^2 = 0.99$, **but** lognormal preferred.
- Cohorts *shrink and elongate* from October to late November, with a clear mid-November rebound.
- Triangle counts mirror the size rebound; sentiment stays mildly positive.
- SIS contagion captures $\sim 56\%$ of cumulative negative volume $\rightarrow$ shape ok, cause unrecoverable.

Next steps

- temporal bipartite mention/retweet graph
- joint positive/negative epidemics
- rerun on the post-March-2020 window for a pre-vs-pandemic diff

Questions?

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